How to Run the NMR

Exam 1 Unknown · CHM 2211 Organic Chemistry II · a step-by-step guide to the virtual NMR room

This guide takes you through every step of recording the NMR spectra of your unknown: making up the sample, putting it into the magnet, locking and shimming, recording ^{1}H, ^{13}C and DEPT-135 spectra, and processing them until they can be read. Keep it open beside the lab. Each step says **what to do**, **what you should see**, and **why** the step matters on a real spectrometer, because the room works the way a real one does.

Before you start

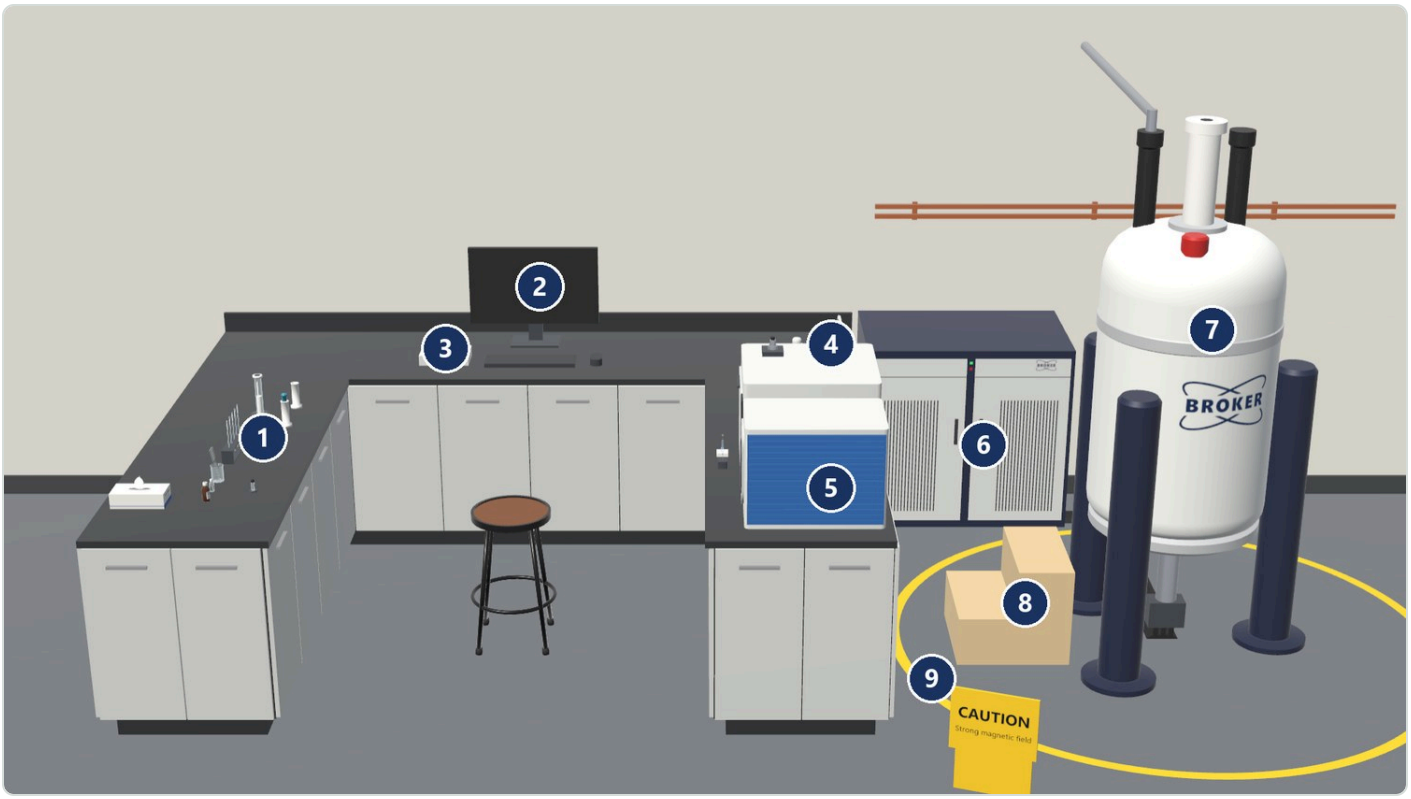
- Allow about 45 minutes for the NMR part of the lab. The IR and the GC-MS take about 15 more.
- The lab saves your work in your browser as you go: you can close the page and come back on the same computer and browser.
- Every step can be done in two ways: in the 3D room (click a thing to pick it up, then click where it goes) or with the buttons in the list of steps beside the room. Use whichever you like; this guide gives both.
- The note under the room always says what to do next, and why, when the room refuses a step.
- The spectra pictured in this guide come from a **practice sample**, not from any of the unknowns.

At a real NMR magnet

- **The magnet is always on.** Its coil is superconducting, kept at −269 °C in liquid helium, and its field never switches off.
- **The 5 gauss line** (the yellow ring on the floor) marks where the field becomes strong enough to matter. Anyone with a pacemaker or another electronic implant stays outside it.
- Leave steel tools, keys, phones, watches and cards with magnetic stripes outside the line. Steel can be pulled into the magnet hard enough to injure someone; phones and cards can be ruined.
- Put nothing into the top of the magnet except a sample tube in its spinner, and only on the lift air.
- If the magnet ever hisses loudly and white fog pours out (a *quench*), leave the room at once: the helium boiling off can push the oxygen out of the air.

THE ROOM

Where everything is

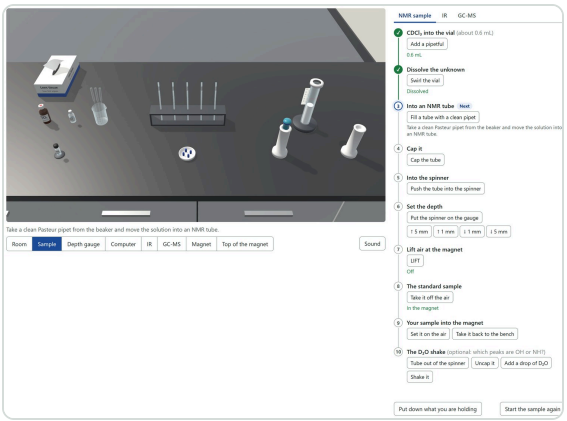


The room as you first see it (the Room view).

1. Making up the sample
2. The spectrometer's computer
3. Keypad: LIFT and the shims
4. The IR (behind the GC-MS)
5. The GC-MS
6. The console (its electronics)
7. The magnet
8. Step stool to the magnet's top
9. The 5 gauss line

How to work in the room

- **Pick up and use:** click a thing to pick it up, then click where it goes. Click empty bench, or press Put down what you are holding, to put it back.
- **Views:** the buttons under the room (Room, Sample, Depth gauge, Computer, Magnet, Top of the magnet…) move the camera. Most steps move it for you.
- **Three lists of steps:** the tabs above the list switch between NMR sample, IR and GC-MS. This guide is about the first.
- **Gold outlines in the room:** the thing under the pointer is outlined in gold, and so is every place the thing in your hand can go.
- **The lights beside the steps:** a blue ring marked Next (with a hint) on the step to do now, a green tick on each step done, a red ! with the reason when the room refuses a step.
- **Sound:** the lift air can be heard. The Sound button turns it off.

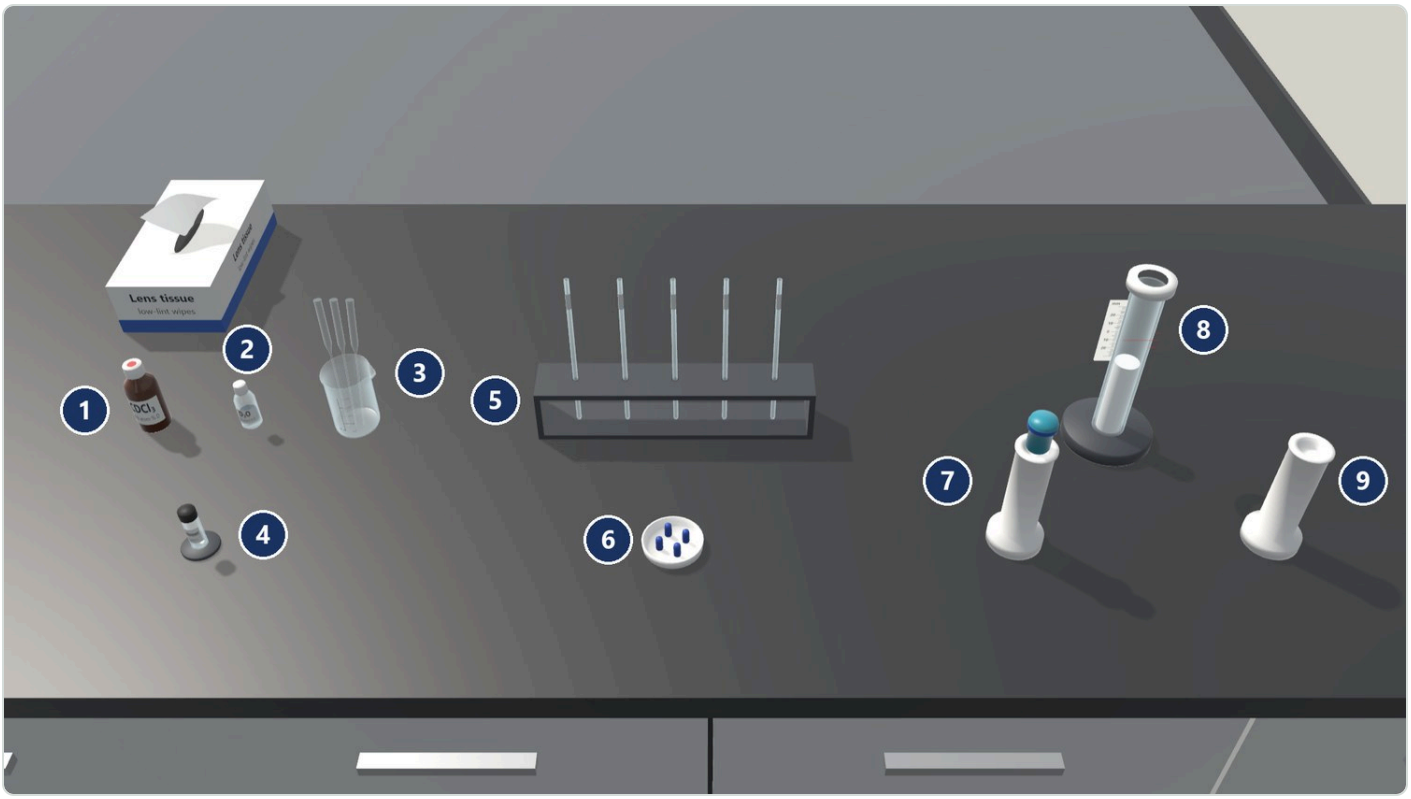


Part 1 of the lab page: the room on the left, the list of steps on the right, each step with its button and its light.

- **The computer:** a click on its screen opens the spectrometer software (part 3 of the page). The screen then shows the spectrometer's status.
- **Starting again:** Start the sample again gives you a clean tube; the spectra you have recorded stay.

PART 1

Make up the sample



The sample bench (the Sample view), left to right in the order you use it: making the solution, filling and capping the tube, the spinner and its depth.

1. $CDCl_3$ (deuterated chloroform)
2. D_2O, for the D_2O shake
3. Clean Pasteur pipets
4. Your unknown, in its vial
5. Clean NMR tubes, in a rack
6. Caps for the tubes
7. The spinner, on its holder
8. The depth gauge
9. The holder for the standard sample

1 Dissolve the unknown in $CDCl_3$: about 0.6 mL

1. Click the **$CDCl_3$ bottle**: you now hold a pipetful, 0.3 mL.
2. Click the **vial** to empty it in.
3. Do it once more, for 0.6 mL in all. (Button: Add a pipetful, twice.)

> **You should see** "0.6 mL" beside the step.

> **Why:** $CDCl_3$ is chloroform whose hydrogens are deuterium. Deuterium gives no peak in a ^1H spectrum, so the solvent does not bury your compound, and the spectrometer uses its signal to hold the field steady (the lock, step 11). Only the trace of ordinary $CHCl_3$ left in it shows, at 7.26 ppm. Use no more than you need: every extra pipetful dilutes the sample and weakens its spectrum.

2 Swirl until it has all dissolved

1. Click the **vial** (button: Swirl the vial). For a solid, swirl until the step says "Dissolved".

> **Why:** solid left in the tube spoils the field round the sample and broadens every line. The room will not let you move solution with solid still in it.

3 Move the solution into an NMR tube

1. Click the **beaker** to take a clean Pasteur pipet.
2. Click the **vial** to draw up the solution.
3. Click a **tube in the rack** to empty the pipet into it. (Button: Fill a tube with a clean pipet.)

> **You should see** about "4.3 cm of solution" beside the step.

> **Why:** an NMR tube is 5 mm across, so 0.6 mL stands about 4.3 cm high. The coil that sends and picks up the signal reads only the middle 1.8 cm, and the liquid must reach well above and below it: with too little, the field is uneven at the liquid's ends and the lines broaden.

4 Cap the tube

1. Click the **dish of caps**, then the **tube**. (Button: Cap the tube.)

> **Why:** $CDCl_3$ evaporates quickly, and a cap keeps the sample from changing while you work.

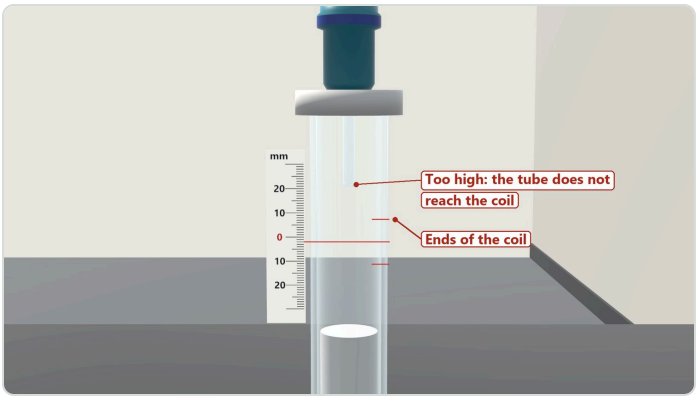
5 Push the tube into the spinner

1. Click your **tube** to pick it up, then the **spinner on its holder**. (Button: Push the tube into the spinner.)

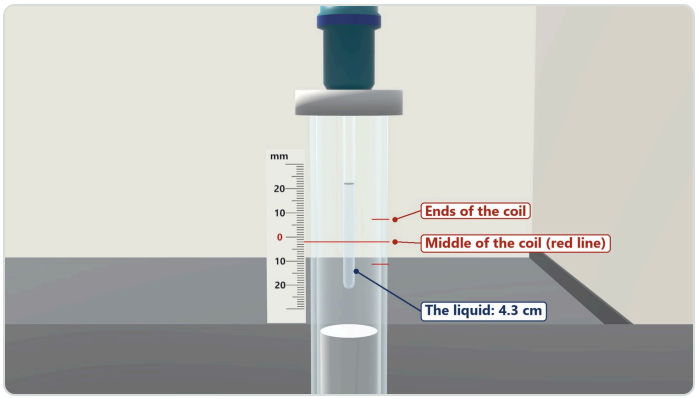
> **Why:** the spinner is the collar that carries the tube into the magnet on a cushion of air, and sets how deep the tube hangs.

6 Set the depth on the gauge

1. Click the **spinner**, then the **depth gauge** (button: Put the spinner on the gauge). The camera moves close to the gauge.
2. Move the tube through the spinner until **the middle of the liquid sits on the long red line**. Drag the tube up or down, or use ↑ 5 mm ↑ 1 mm ↓ 1 mm ↓ 5 mm.
3. Check that the liquid covers **both short red marks** (the ends of the coil).



As the tube first goes in: far too high.



Set right: the liquid's middle on the red line, both ends of the coil covered.

> **Why:** the red line is the middle of the coil inside the probe. The spectrometer reads only the part of the sample inside the coil, and the field is most even there. If the liquid's middle is more than about 2.5 mm off, or the liquid does not cover the coil, the lock is weak and the lines broaden, however well you shim. The gauge's scale is in millimetres above and below the coil's middle.

PART 2

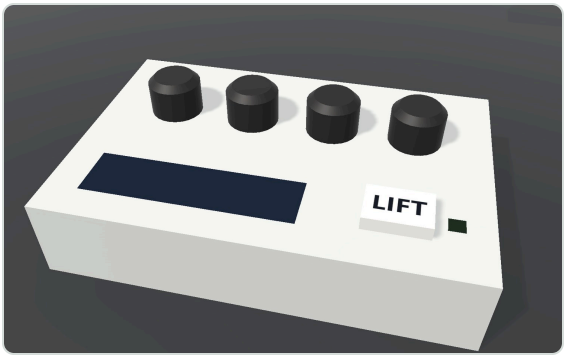
Put the sample into the magnet

The sample is never dropped into the magnet. A stream of air, the **lift air**, rises up the magnet's bore and holds a spinner floating at the top; turned off, it lets the spinner sink gently down onto the probe. There is always a sample in the magnet when nobody is using it: the **standard sample**, which must come out first.

7 Turn the lift air on

1. Press the **LIFT** key on the keypad by the computer, or the LIFT button in the list of steps.

> **You should see** the lamp beside LIFT light and the step say "On", and hear the air rush in. After a few seconds the standard sample (red cap, red band on its spinner) rises and floats at the top of the magnet.



The keypad by the computer: the four shim knobs, the LIFT key and its lamp.

8 Take the standard sample off the air

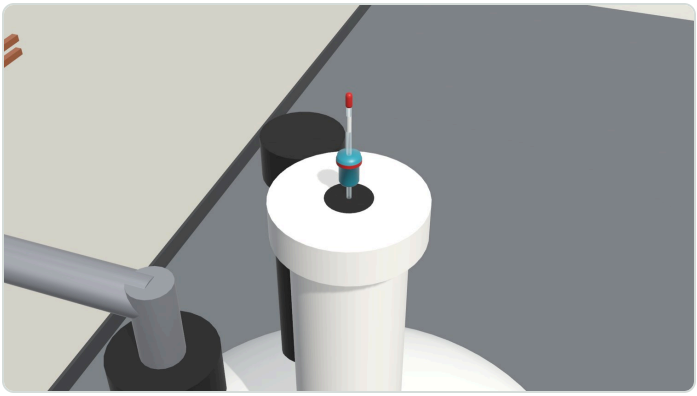
1. Button: Take it off the air. It goes to its own holder on the sample bench.

> The room will not let you put your sample on the air while the standard is still there, or while the air is off: "*The lift air is off... the sample would fall straight down onto the probe.*"

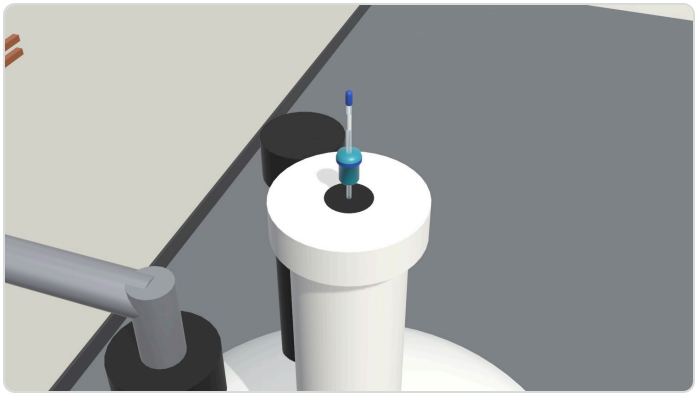
9 Set your sample on the air

1. Click the **spinner** on the gauge to pick it up.
2. Click the **magnet** to climb the step stool, then the **top of the magnet** (button: Set it on the air).

> **You should see** your sample floating at the top in its turquoise spinner, with its blue cap showing.



Step 7: lift air on, and the standard sample (red cap) floats up out of the magnet.



Step 9: your sample (blue cap) on the air, before the air goes off.

10 Turn the lift air off

1. Press **LIFT** again.

> **You should see** the sample sink out of sight and hear the air fall away. The step says "In the magnet", and the console's status (part 3 of the page) says "Sample in the magnet".

The last item in the list, the D_2O shake, is optional: part 6 of this guide.

PART 3

Lock and shim

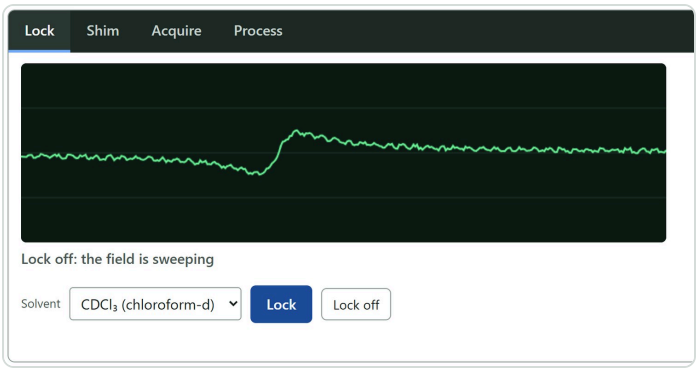
The rest happens in **part 3 of the lab page, "The spectrometer"** (a click on the computer's screen in the room takes you there): the spectrometer's software, with four tabs, Lock, Shim, Acquire and Process. The status at the top right always says whether the sample is in the magnet and whether the spectrometer is locked.

11 Lock on the solvent's deuterium

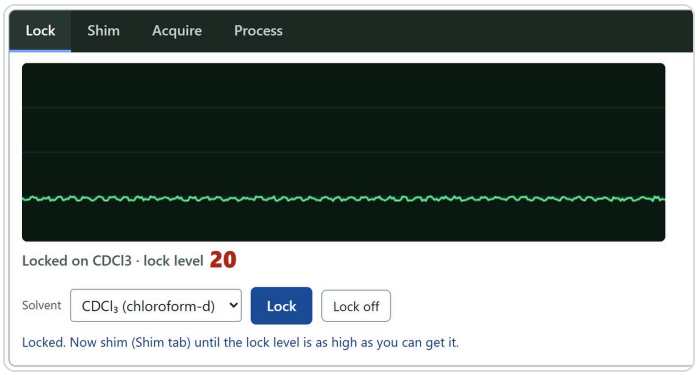
1. Open the Lock tab.
2. Choose the solvent your sample is in, **$CDCl_3$ (chloroform-d)**, and press Lock.

> **You should see** the display change from a moving wiggle ("Lock off: the field is sweeping") to "Searching for the deuterium signal…", then, after about two seconds, a flat line and **"Locked on CDCl3 · lock level …"**. The level will be low at first, often 20 or so.

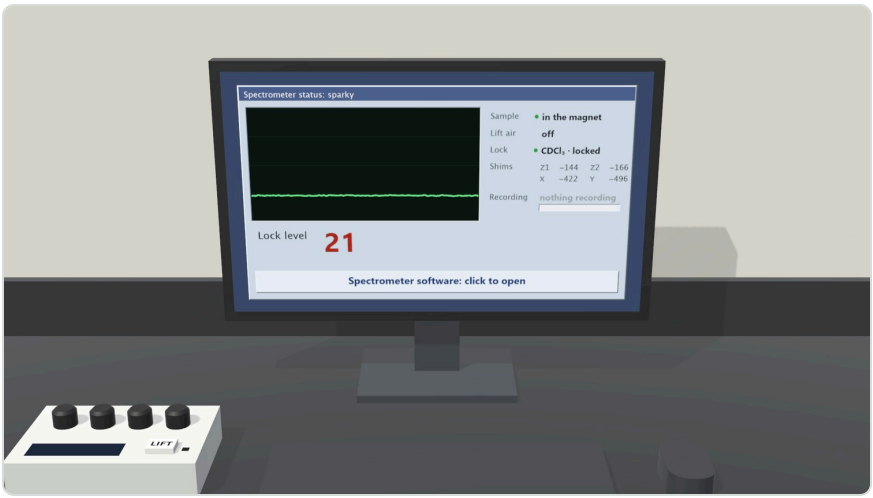
> **Why:** the field drifts very slightly all the time. Locked, the spectrometer watches the solvent's deuterium signal and corrects the field to keep that signal still, so the lines of your compound stay put while scans are added up. Unlocked, the lines smear and broaden. Choose the wrong solvent and there is nothing to lock on: *"No lock signal found… Is that the solvent your sample is in?"*



Before locking: the field sweeps and the deuterium signal passes by.



Locked: a flat line. Its height is the lock level.



The computer's screen in the room (the Computer view) follows the spectrometer once the software is open: here the sample is in the magnet and locked, lock level 20.

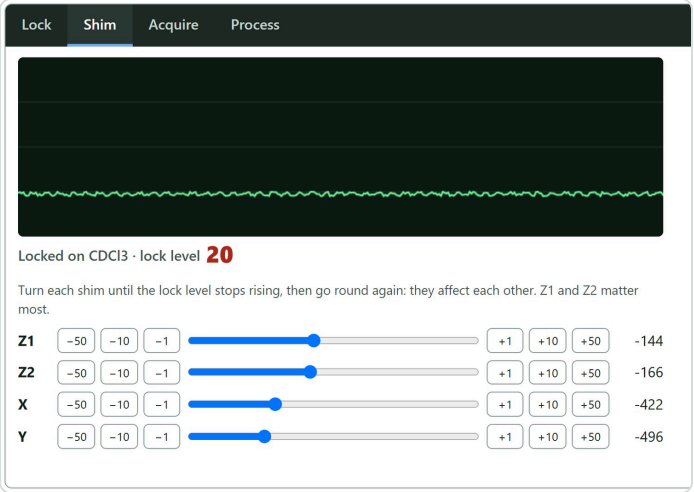
12 Shim until the lock level stops rising

1. Open the Shim tab. Keep an eye on the lock level above the shims.
2. Start with **Z1**. Press +10. If the level rises, keep pressing it until the level starts to fall, then go back one step. If it falls at once, go the other way with −10.
3. Fine-tune Z1 the same way with +1 and −1.
4. Do the same with **Z2**, then **X**, then **Y**. Use ±50 when a shim is far out and the level hardly moves.
5. Go round all four again. The shims affect one another, so the second round usually finds a little more. Stop when no shim raises the level any further.

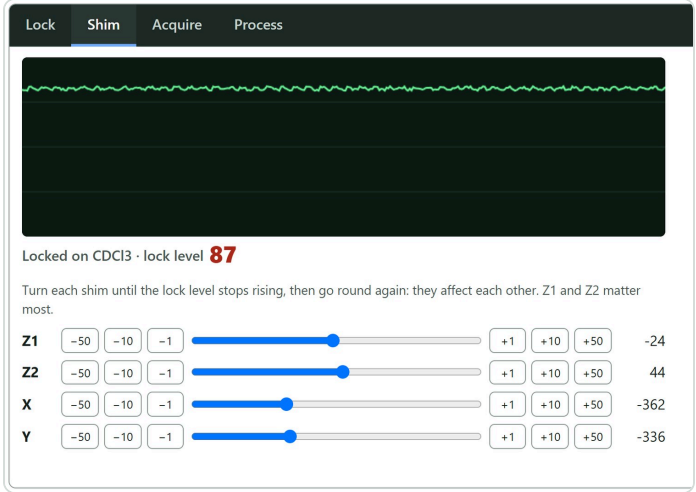
You should see the level climb into the 80s for a well-made sample at the right depth; close to 90 is as good as it gets. There are no correct numbers to copy: every spectrometer, and every student's magnet here, has its own best settings. (The four knobs on the keypad by the computer are these same shims; here you set them in the Shim tab.)

Why: the shims are small extra coils that make the field the same all through the sample. The more even the field, the narrower the lines, and the taller and steadier the lock signal, so the lock level is the measure to shim by. Z1 matters most, then Z2; X and Y less. A Z1 left out of adjustment shows later as a lopsided "shoulder" on every peak.

If the level stays well below 80 however long you shim, the sample itself is the problem: in the 60s, the liquid is probably off the coil's middle; below about 50, it probably does not cover the coil. Take the sample out (LIFT on, Take it back to the bench), set the depth again, and put it back.



Before shimming: lock level about 20.



After two rounds of all four shims: lock level 87.

PART 4

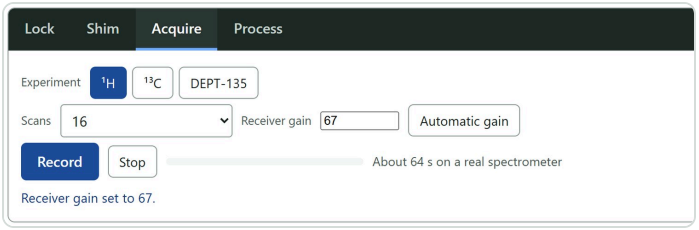
Record the spectra

13 Record a ^{1}H spectrum

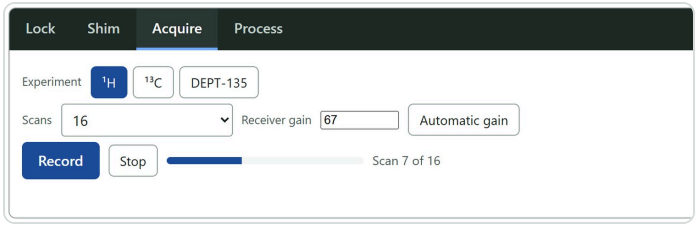
1. Open the Acquire tab and choose ^{1}H.
2. Set **Scans** to **16**.
3. Press Automatic gain.
4. Press Record.

> **You should see** the bar fill and "Scan 3 of 16…" count up. When it finishes, the page says "Recorded: ^{1}H, 16 scans." and opens the Process tab. It takes about 16 seconds here; the page shows how long it would take on a real spectrometer (about a minute).

> **Why:** each scan is one pulse and one recording of the signal, the FID. Scans are added together: the signal grows with the number of scans and the noise only with its square root, so four times the scans gives a spectrum twice as clean. The **receiver gain** sets how much the signal is amplified before it is turned into numbers. Too high, and the start of the FID is cut off ("The receiver gain was too high: the FID was clipped"), which wrecks the spectrum; too low, and the spectrum is noisier than it need be. Automatic gain finds the right setting.



Ready: ^{1}H, 16 scans, gain set automatically.



Recording, scan by scan.

14 Record a ^{13}C spectrum

1. Choose ^{13}C, set **Scans** to **256** or more, press Automatic gain, then Record.

> **Why so many scans:** only about 1 carbon atom in 100 is ^{13}C, and its signal is far weaker than a proton's, so it takes hundreds of scans to rise clear of the noise. Carbons with no hydrogens on them are weaker still. If small peaks are lost in the noise, record again with 1024 scans.

15 Record a DEPT-135 spectrum

1. Choose DEPT-135, **256** scans, Automatic gain, Record.

> **Why:** DEPT-135 shows each carbon according to how many hydrogens it carries: CH and CH_3 point up, CH_2 point down, and carbons with no hydrogen do not appear. Read beside the ^{13}C spectrum, it sorts every carbon (step 22).

Recording works only with the sample in the magnet. If you record without locking, the page warns you: the lines will be broader.

PART 5

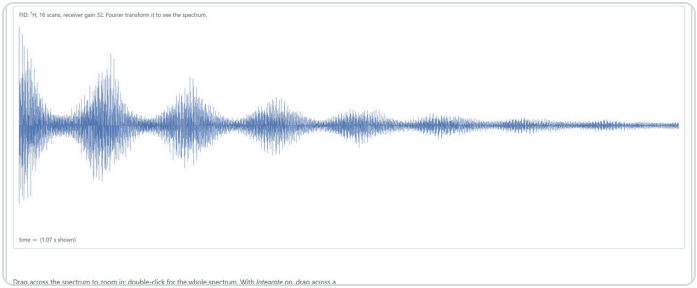
Process each spectrum

A spectrometer records a signal that dies away in time, the FID. Processing turns it into a spectrum you can read. In the Process tab, choose the spectrum with the buttons at the top (^{1}H , ^{13}C , DEPT-135), then work along the toolbar from left to right. Do these steps for **each** spectrum. Each one remembers its own processing.

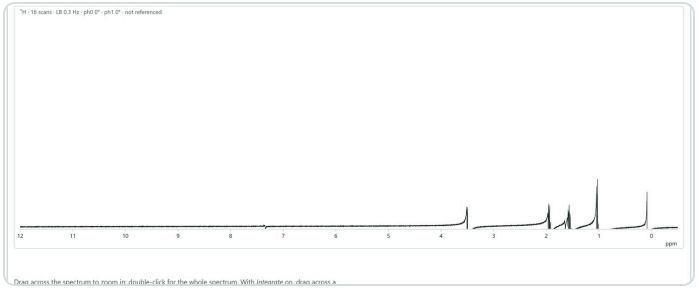
16 Fourier transform

1. Press Fourier transform . Leave **LB** as it is: 0.3 Hz for ^{1}H, 1 Hz for ^{13}C and DEPT.

> **Why:** the FID is all the frequencies of your sample's nuclei sounding together; the Fourier transform sorts them out into peaks. LB (line broadening) smooths the noise a little at the cost of slightly broader lines; press Fourier transform again after changing it.



The FID, as recorded.



After the Fourier transform, before phasing: the peaks lean and dip, and the baseline rolls.

17 Phase the spectrum

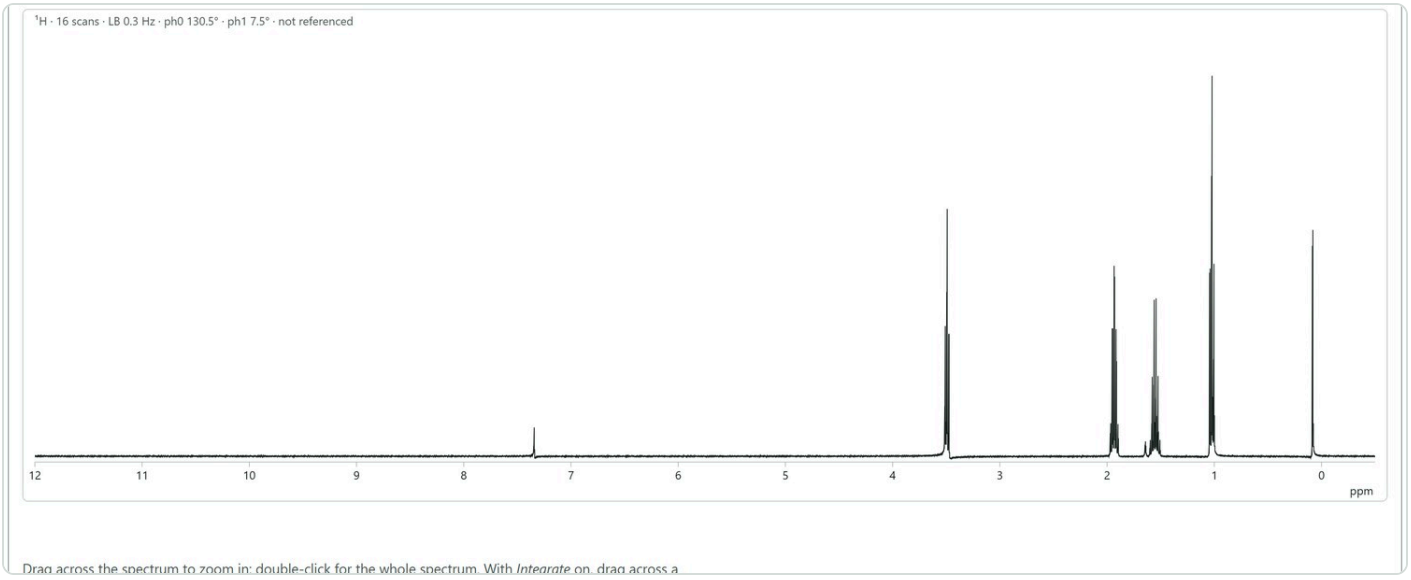
1. Press Automatic phase .
2. Look closely (drag across a group of peaks to zoom in, step 21): every peak should stand **upright and symmetric**, with flat baseline on both sides.
3. If not, adjust by hand with small moves: **ph0** changes every peak alike; **ph1** changes peaks more the further they are across the spectrum. Fix the peaks at one end with ph0 first, then those at the other end with ph1.

> **Why:** the spectrometer starts recording a moment after the pulse, and every frequency has drifted out of step by a different amount; phasing brings them back in step so every peak is a clean, upright, measurable shape. For **DEPT-135**, phase the ^{13}C spectrum first: Automatic phase then gives the DEPT the same correction, and its CH_2 peaks point down.

18 Correct the baseline

1. Press Baseline correction .

> **Why:** a baseline that is not flat and at zero throws off integrals and hides small peaks.



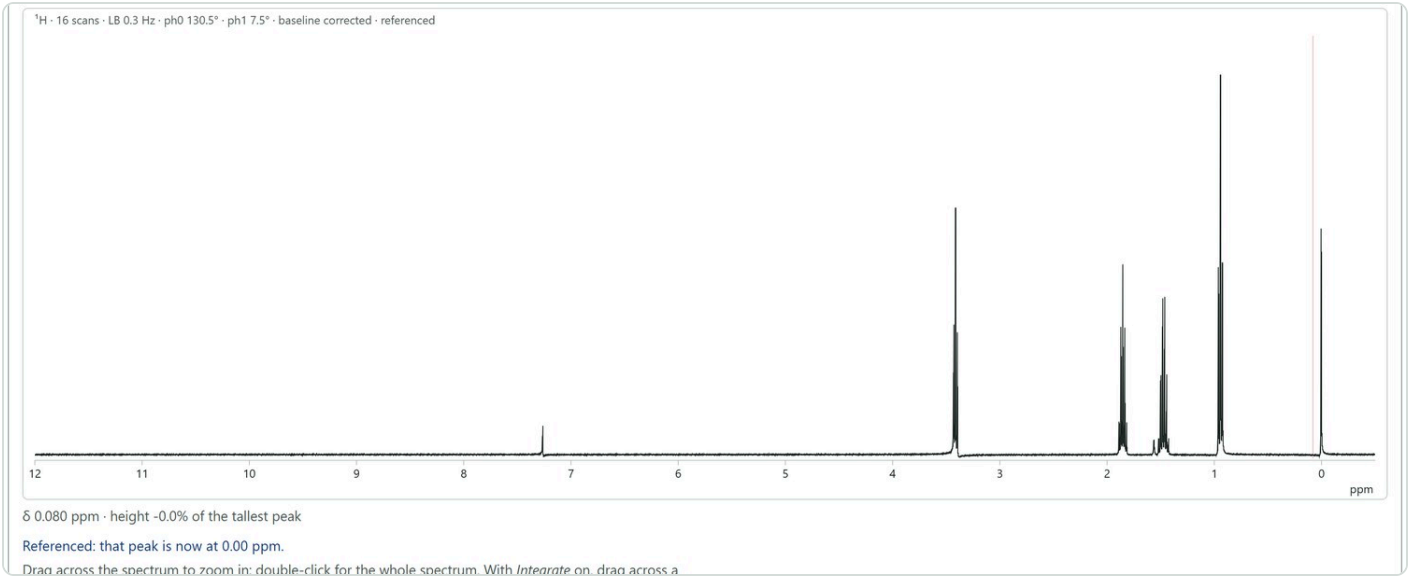
Phased: every peak upright. Not yet referenced: TMS, the small peak at the far right, reads 0.08 ppm.

19 Reference the spectrum

1. Under **Reference**, choose **TMS = 0.00 ppm**.
2. Press Click its peak, then click on (or right next to) the **TMS peak**, the small sharp peak at the far right, near 0 ppm.

> **You should see** "Referenced: that peak is now at 0.00 ppm." and the line at the top of the spectrum change from "not referenced" to "referenced".

> **Why:** chemical shifts are measured from TMS (tetramethylsilane, added to the solvent), which is set at exactly 0. An unreferenced spectrum here reads up to about 0.15 ppm off, enough to make every shift you report wrong. $CHCl_3$ at 7.26 ppm (^1H) or the middle line of the $CDCl_3$ triplet at 77.16 ppm (^{13}C) can be used instead. Reference **every** spectrum, DEPT-135 included.



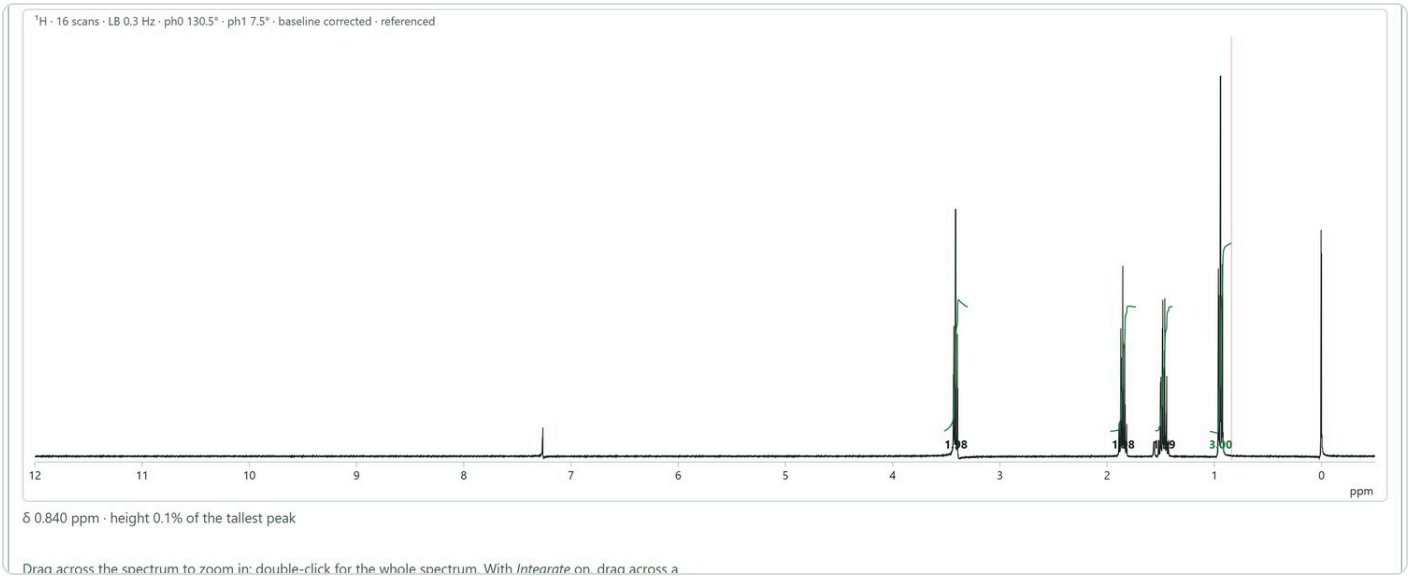
Baseline corrected and referenced: TMS now at 0.00 ppm, and every other peak moved with it.

20 Integrate the ^{1}H spectrum

1. Press Integrate to turn it on.
2. Drag across each group of peaks, one group at a time, from a little before it to a little after it. A green running total and a number appear for each.
3. Click the number of a group whose number of H you are sure of (a CH_3 singlet, say), type that number in "This integral stands for … H", and press Set. All the others scale to match.
4. Press Integrate again to turn it off. Clear integrals starts again; a group's number also offers Remove it.

> **You should see** numbers close to whole numbers. For the practice sample: 2, 2, 2 and 3. They should add up to the number of H in your formula.

> **Why:** the area of a group of peaks is proportional to the number of hydrogens that give it. Leave the solvent, water and TMS peaks out of every region.

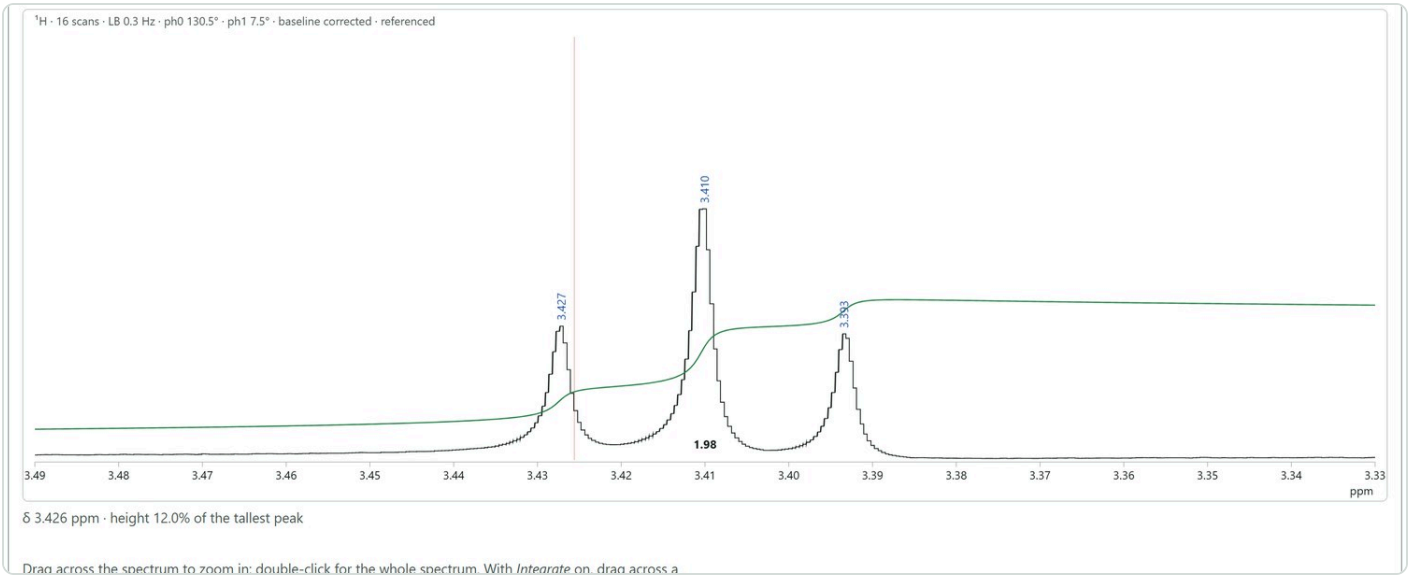


Four groups integrated, the triplet near 0.9 ppm set to 3 H.

21 Pick peaks, zoom in and measure coupling constants

1. Press Pick peaks: each peak is labelled with its shift. Move the **above … %** slider to label smaller peaks, or fewer.
2. **Zoom:** drag across the part you want; double-click, or Full view, for the whole spectrum. Taller and Shorter (or the mouse wheel) change the height.
3. The line under the spectrum always gives **δ** (ppm) under the pointer.
4. For a multiplet, count its lines and compare their heights (1 : 2 : 1 is a triplet, 1 : 3 : 3 : 1 a quartet), take its shift as the middle, and measure **J**: the gap between neighbouring lines in ppm, times 400. At 400 MHz, 0.01 ppm is 4 Hz.

> **Example** (the practice sample, below): lines at 3.427, 3.410 and 3.393 ppm, heights 1 : 2 : 1, so a **triplet at 3.41 ppm**, with J = 0.017 × 400 = **6.8 Hz**.

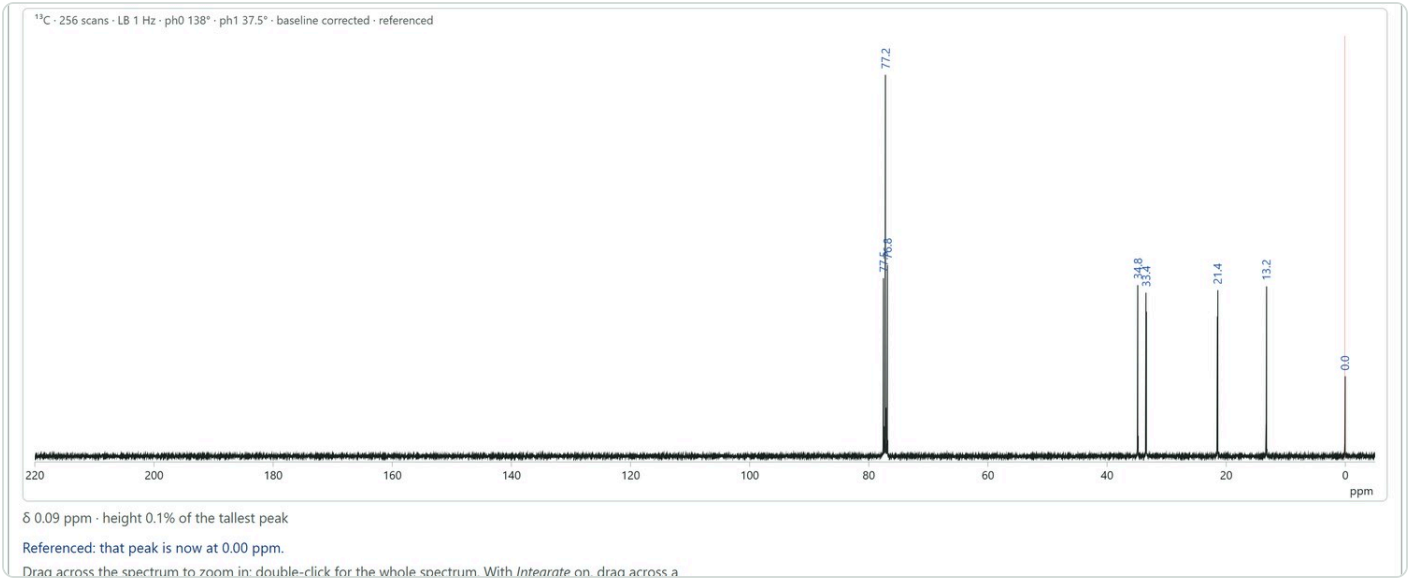


Zoomed in on one multiplet, peaks picked, with its integral running across it.

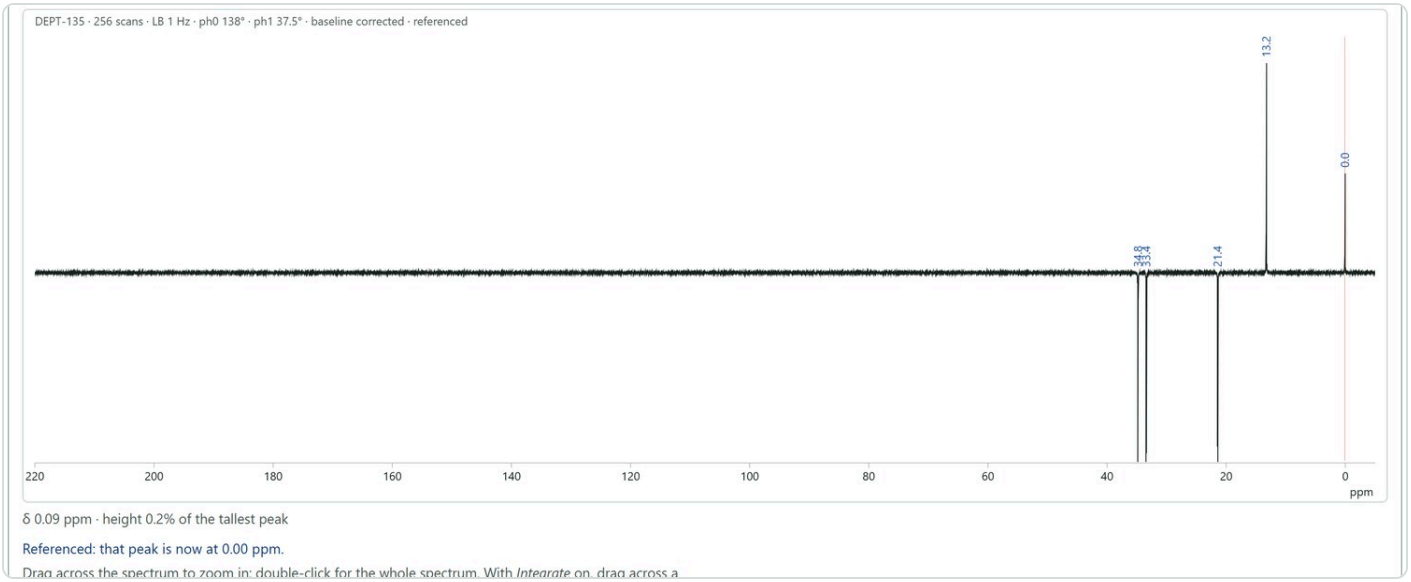
22 Read the ^{13}C and DEPT-135 spectra together

1. Process both (steps 16 to 19), then Pick peaks on each.
2. Every ^{13}C peak is one kind of carbon. Look for it in the DEPT-135: **up** = CH or CH_3; **down** = CH_2; **missing** = a carbon with no H.
3. Leave out the $CDCl_3$ triplet at 77.16 ppm and TMS at 0.

> **Why:** ^{13}C spectra are recorded with the protons decoupled, so every carbon is a single line; their heights do not count carbons, unlike ^{1}H integrals.



^{13}C: four carbons (34.8, 33.4, 21.4 and 13.2), the $CDCl_3$ triplet (77.2) and TMS (0.0).



DEPT-135: the three carbons at 34.8, 33.4 and 21.4 point down (CH_2); 13.2 points up (CH_3), as does TMS. No solvent peak.

23 Enter what you read

1. In **part 4 of the page**, fill Table 1 (^{1}H: δ to two decimals, splitting, J, number of H, which H) and Table 2 (^{13}C: δ to one decimal, what DEPT-135 shows, which C).
2. The page checks each row as you type: a tick when a box is right, a red mark and a hint when it is not. "Which H" and "Which C" are your own reasoning and are not checked.

> **Hints you may see:** "*…about 0.13 ppm out: reference the spectrum (TMS at 0.00)…*" (step 19); "*That is the $CHCl_3$ left in the $CDCl_3$*" (a solvent peak); "*Zoom in: these lines can be counted*" (name the splitting); "*Check the integration*" (step 20).

PART 6 · OPTIONAL

The D_2O shake: which peaks are OH or NH?

The hydrogen of an OH or NH group trades places with deuterium from D_2O. Once it is deuterium it gives no ^{1}H peak, so after a shake with D_2O the OH or NH peak shrinks or disappears, and a new peak of HOD appears near 4.7 ppm. Do this after you have recorded and processed your first ^{1}H spectrum.

24 Take your sample out

1. Turn the lift air on (LIFT); your sample floats up.
2. Take it back to the bench: the spinner goes back on its holder.
3. Tube out of the spinner, then Uncap it.

25 Add D_2O and shake

1. Add a drop of D_2O (or click the D_2O bottle, then your tube).
2. Cap the tube again (dish of caps, then the tube).
3. Shake it. Without the shake only half of the OH or NH hydrogens trade places.

26 Put it back and record again

1. Back into the spinner, set the depth on the gauge again (step 6), and put it into the magnet (steps 7 to 10; the standard is already out).
2. Lock again (step 11) and check the shims (step 12): a sample that has been out and back may need a touch.
3. Record a ^{1}H spectrum (step 13). It appears as a separate spectrum, `$^{1}H$ after $D_2O$`, so you can compare the two. Process it the same way.

> **Compare** the two ^{1}H spectra: a peak that has shrunk or vanished is an OH or NH; the new peak near 4.7 ppm is HOD, not your compound.

WHEN SOMETHING GOES WRONG

Troubleshooting

What you see	Why	What to do
"Some of the unknown is still solid."	A solid has not fully dissolved.	Swirl the vial (step 2) before filling the tube.
"The lift air is off..." when setting the sample on the magnet	Without air the sample would drop onto the probe.	Press LIFT first (step 7).
"Take the standard sample off the air first."	Only one spinner can float at the top.	Step 8.
"Put your sample in the magnet first (part 1)."	Lock and Record need a sample in the magnet.	Finish part 2 of this guide: turn the air off so the sample goes down.
"No lock signal found for ..."	The wrong solvent was chosen.	Choose $CDCl_3$ and lock again.
Lock level stays well below 80 however you shim	The liquid's middle is off the coil's middle, or the liquid does not cover the coil.	Take the sample out and set the depth again (step 6): middle on the red line, both short marks covered.
Broad, lumpy peaks, or a shoulder on every peak	The shims, Z1 above all, are off; or the spectrum was recorded unlocked.	Lock, shim again (step 12), record again.
"The receiver gain was too high: the FID was clipped."	The signal overflowed the digitizer.	Press Automatic gain, record again.
Peaks dip below the baseline on one side, or point down	The spectrum is not phased.	Automatic phase, then ph0 and ph1 by hand (step 17).
Your shifts are all out by about 0.1 ppm; the page says "reference the spectrum"	The spectrum was not referenced.	Step 19, then read the shifts again.
Integrals are not near whole numbers, or do not add up	A region includes a solvent peak, or misses part of a multiplet, or no integral has been set.	Clear integrals and drag again; set one group you are sure of (step 20).
Peaks at 7.26, 1.56 or 0.00 ppm (^{1}H); 77.16 or 0.00 ppm (^{13}C)	$CHCl_3$ in the solvent, a trace of water, TMS; the $CDCl_3$ triplet.	Leave them out of the tables.
Small ^{13}C peaks lost in the noise	Not enough scans.	Record again with 1024 scans: four times the scans halves the noise.
A DEPT-135 with no peaks pointing down	Either there is no CH_2, or the DEPT is phased upside down.	Phase the ^{13}C first, then press Automatic phase on the DEPT again.

Quick reference

Make up the sample

1. $CDCl_3$ into the vial, two pipetfuls (0.6 mL).
2. Swirl until dissolved.
3. Clean pipet: vial → NMR tube (about 4.3 cm).
4. Cap it.
5. Tube into the spinner.
6. Gauge: liquid's middle on the red line, both short marks covered.

Into the magnet

7. LIFT on: the standard floats up.
8. Standard off the air.
9. Your sample on the air.
10. LIFT off: "In the magnet".

Lock and shim

11. Lock tab: $CDCl_3$, Lock.
12. Shim tab: Z1, Z2, X, Y, twice round, until the level stops rising (80s).

Record

13. ^1H: 16 scans, Automatic gain, Record.
14. ^{13}C: 256+ scans, Automatic gain, Record.
15. DEPT-135: 256 scans, Automatic gain, Record.

Process (each spectrum)

16. Fourier transform (LB as given).
17. Automatic phase; check, adjust ph0 and ph1.
18. Baseline correction.
19. Reference: TMS = 0.00, Click its peak.
20. ^1H: Integrate each group; set one you are sure of.
21. Pick peaks; zoom; J = Δppm × 400 Hz.
22. ^{13}C with DEPT: up CH/CH_3, down CH_2, missing C.
23. Fill Tables 1 and 2 (part 4 of the page).

Optional: D_2O shake

24. Out, uncap, drop of D_2O, cap, shake, back in, depth, magnet, lock, shim, ^1H again: compare.